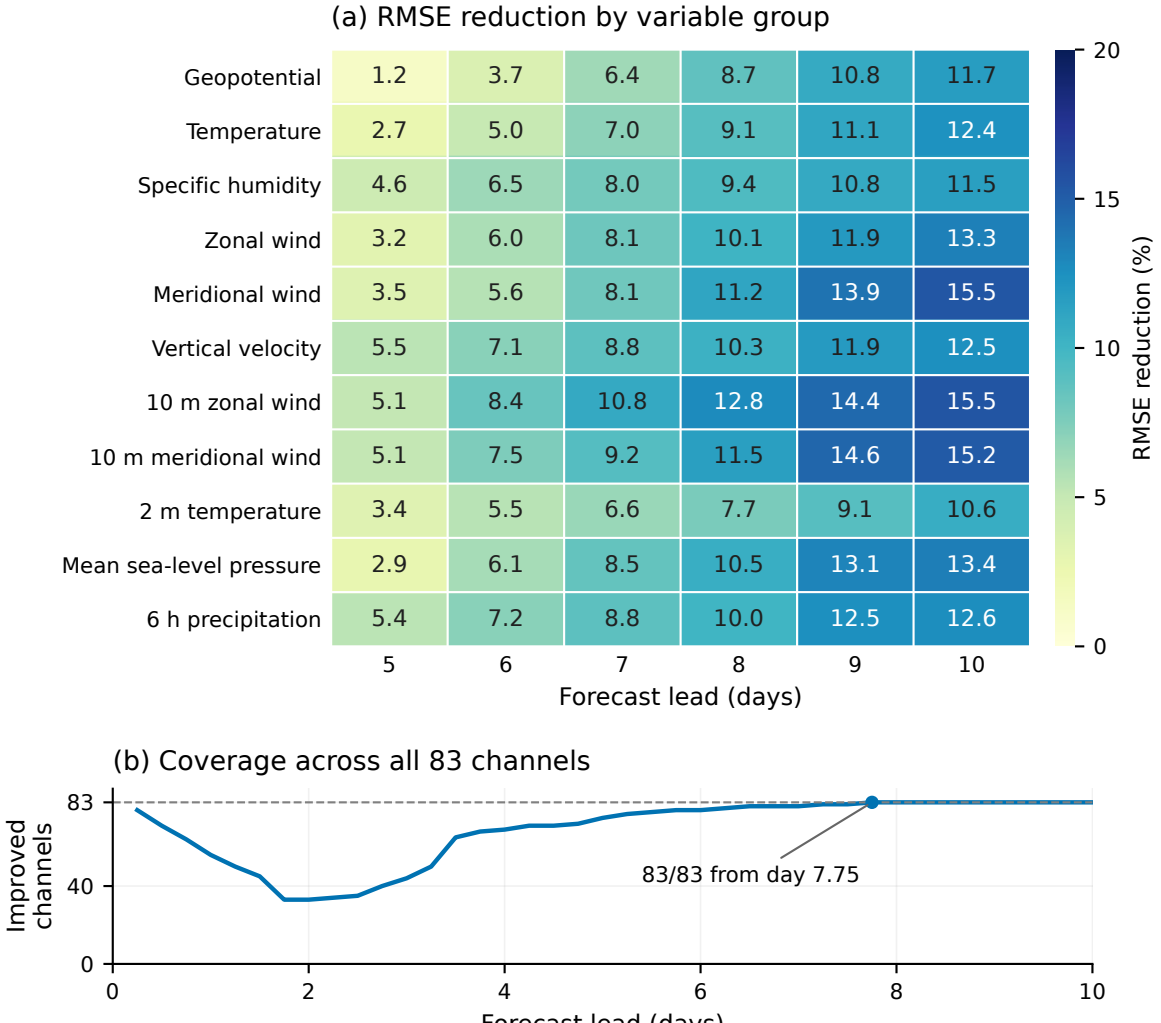
(a) RMSE reduction by variable group
Geopotential 1.2 3.7 6.4 8.7 10.8 11.7
Temperature 2.7 5.0 7.0 9.1 11.1 12.4
Specific humidity 4.6 6.5 8.0 9.4 10.8 11.5
Zonal wind 3.2 6.0 8.1 10.1 11.9 13.3
Meridional wind 3.5 5.6 8.1 11.2 13.9 15.5
Vertical velocity 5.5 7.1 8.8 10.3 11.9 12.5
10 m zonal wind 5.1 8.4 10.8 12.8 14.4 15.5
10 m meridional wind 5.1 7.5 9.2 11.5 14.6 15.2
2 m temperature 3.4 5.5 6.6 7.7 9.1 10.6
Mean sea-level pressure 2.9 6.1 8.5 10.5 13.1 13.4
6 h precipitation 5.4 7.2 8.8 10.0 12.5 12.6
5 6 7 8 9 10
Forecast lead (days)
RMSE reduction (%)
0 5 10 15 20
(b) Coverage across all 83 channels
Improved channels
83
40
0
83/83 from day 7.75
0 2 4 6 8 10
Forecast lead (days)